



School of Biological Sciences

Corticosteroid Receptor–Genome Interactions Under Circadian and Stress Conditions in the Rat Hippocampus: A ChIP-seq Differential Binding Analysis

By
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Abstract

Glucocorticoids act through mineralocorticoid receptors (MR) and glucocorticoid receptors (GR) to shape hippocampal function across the circadian cycle and during acute stress. I re-analysed public ChIP seq from adult rat hippocampus to test whether forced swim stress (FS) leaves a distinct genomic footprint when the system is already at the late afternoon circadian peak, BLPM. Genome wide binding of GR, MR and the histone mark H3K9acS10p (S10) was profiled at BLAM, BLPM and 30 minutes after FS. Peaks were called with MACS2 and differential binding was assessed with DiffBind and DESeq2 on consensus peak sets. Peaks were annotated with ChIPseeker and pathway context was assessed with Ingenuity Pathway Analysis. The primary contrast, FS versus BLPM, showed a selective GR increase with 181 peaks at $FDR \leq 0.05$ and 285 at $FDR \leq 0.1$ and a mean log2 fold change of about 1.07. MR changed at far fewer sites, 11 to 14, and S10 differed at only 9 to 22 loci with near zero effect sizes. In BLAM traversing contrasts both receptors showed large gains, while S10 changes were modest. PCA and heatmaps separated states most clearly for GR. FKBP5 appeared whenever BLAM was the comparator but not in GR FS versus BLPM, consistent with a sentinel of glucocorticoid availability. Condition wise overlaps showed an S10 rich landscape that expanded from BLAM to BLPM, with the $GR \cap MR \cap S10$ set rising from 126 to 218 and remaining 218 after FS. Pathway analysis highlighted circadian and neuroplasticity programmes including GPCR, cAMP, ERK, CREB, PI3K AKT, cytoskeleton and axonal guidance.

Author's Declaration

I declare that the work in this dissertation was carried out in accordance with the Regulations of the University of Bristol. The work is original, except where indicated by special reference in the text, and no part of the dissertation has been submitted for any other academic award. Any views expressed in the dissertation are those of the author.

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Keywords

Hippocampus; Corticosteroid receptors; H3K9acS10p; ChIP-seq; Forced-swim stress (FS); Circadian rhythm.

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1. Introduction

Glucocorticoid (GC) hormones are key regulators of brain function, supporting behavioural adaptation, metabolic balance, and the acute stress response (1). In rodents, corticosterone secretion is controlled by the hypothalamic–pituitary–adrenal (HPA) axis and follows a circadian rhythm that peaks at the onset of the active phase (2). Acute stress rides on top of this rhythm, producing a rapid, transient surge in hormone and broad transcriptional effects in GC-sensitive tissues such as the hippocampus (3).

Two nuclear receptors mediate GC action in the brain, the mineralocorticoid receptor (MR) and the glucocorticoid receptor (GR) (4). These receptors differ in affinity and operating range (5). MR has high affinity and is substantially occupied even at basal hormone levels, contributing to neuronal excitability and synaptic stability. GR has lower affinity and is increasingly recruited as hormone concentrations rise at the circadian peak or during stress (6). Once activated, both receptors translocate to the nucleus, bind glucocorticoid response elements (GREs), and cooperate with other factors to regulate genes involved in mood, cognition, metabolism, and immune balance (7).

The hippocampus expresses both receptors at high levels and is a major site where GC signals shape memory and provide feedback to the HPA axis (8). MR and GR often co-regulate genomic programmes relevant to plasticity and behaviour, yet the extent of their functional overlap is context-dependent. Early ChIP assays at single loci such as, *Fkbp5*, *Per1*, *Sgk1*, showed state-dependent variation in receptor binding but could not reveal genome-wide patterns or the chromatin environment that enables receptor access (9, 10). Previous work mapped MR and GR binding across three physiological states in rat hippocampus, BLAM, BLPM (circadian peak) and 30 min after forced-swim stress (FS). They reported widespread receptor engagement with overlapping yet distinct site sets and pathway signatures linked to neuronal plasticity, cytoskeletal control and behavioural adaptation (1). They also showed that both BLPM and FS, glucocorticoid-high states, display expanded receptor occupancy relative to BLAM. However, most analyses focused on contrasts that traverse the hormone trough (BLPM vs BLAM; FS vs BLAM). A direct comparison of FS with BLPM, the two GC-high states, was not the central analysis, leaving open how stress modifies receptor binding on a circadian pre-engaged genome.

This missing comparison is important. FS and BLPM share high hormone availability but differ in origin and neuromodulatory context (11). FS is a rapid, stress-evoked surge; BLPM reflects the regular circadian peak at the start of activity (12). Comparing each state to BLAM mixes two influences, hormone availability and state-specific cues. Directly contrasting FS with BLPM isolates the stress-specific layer added to a genome already engaged by the circadian rise. This frames a test of circadian gating, the idea that time of day sets a baseline of receptor occupancy and cofactor availability (13, 14, 15), which then shapes what

additional binding and signalling stress can elicit at 30 min. If gating is strong, the incremental FS signal on a circadian background should be selective and smaller than changes observed when moving from GC-low to GC-high (16). If gating is weak, FS should still recruit a broad, GR-centred programme despite prior circadian engagement (17) (18). Chromatin state is likely to contribute to this gating. Histone modifications and accessibility together shape when, and how strongly, transcription factors can act, thereby tuning the speed and magnitude of transcriptional responses (19). S10 is associated with permissive, active chromatin in neurones and rises after stress and learning. Accordingly, time-of-day or context-dependent shifts in these chromatin features could gate GR recruitment and output (20).

The hippocampus is a suitable system in which to test these ideas. It shows pronounced circadian and stress-responsive programmes with behavioural consequences and contains cell populations in which both receptors operate (21). MR has been linked to appraisal and steady-state control across the day (22, 23); GR is associated with larger, short-lived responses during stress and at the circadian peak (24). At a genome-wide level, both receptors occupy thousands of sites near genes involved in neurite growth, axonal guidance, cytoskeletal dynamics and myelination (25). What is not yet defined is how MR and GR binding differ between FS and BLPM, and whether any differences at this acute timepoint are mirrored by S10 at the same genomic regions.

The working model is that circadian phase gates receptor action by pre-establishing a permissive chromatin and co-factor environment on which acute stress can build (26, 27). Consistent with this, MR, given its higher affinity, occupies many targets across the day and shows its largest shifts when crossing the morning trough, whereas GR, with lower affinity, displays a broader dynamic range, gaining thousands of sites from trough to peak yet retaining inducible capacity at the circadian peak (28). If gating is strong, a direct FS vs BLPM contrast should yield a focused, GR-centred increment rather than wholesale rewiring. MR should add little at peak because much of its programme is already engaged. For S10 at 30 min, we expect similar levels between these two GC-high states, either because any early pulse has subsided or because the mark approaches a permissive ceiling by late afternoon. Functionally, peak-anchored gene sets derived from GR/MR binding should re-identify synaptic-plasticity modules (neuromodulatory GPCR→cAMP/ERK→CREB), axon-guidance pathways and RHO-dependent cytoskeletal control with only a modest circadian component; FS vs BLPM should recover a subset of this programme with attenuated magnitudes. The analyses below test these phase-aware predictions (1).

1.5.1 Rationale

Directly comparing FS with BLPM isolates stress-specific receptor binding on a circadian, GC-high background. Adding S10 tests whether a permissive chromatin state explains any differences in MR/GR recruitment between these two conditions.

1.5.2 Hypotheses

1. GR shows a selective increase in binding in FS vs BLPM, whereas MR changes are smaller and more stable.
2. Differentially bound regions in FS vs BLPM are frequently S10-enriched (including known GC-responsive loci such as Fkbp5).
3. S10 behaves chiefly as a permissive mark at 30 min, with limited state-specific change between FS and BLPM.

1.5.3 Aims and Objectives

Aims

1. To determine how MR and GR binding differs between FS and BLPM in rat hippocampus.
2. To assess whether S10 tracks those differences and to place findings in functional context.

Objectives

1. Define consensus MR, GR and S10 peak sets across BLAM/BLPM/FS and quantify FS vs BLPM differential binding.
2. Annotate differentially bound sites to nearby genes and examine enrichment of S10 at those loci.
3. Use pathway analysis on gene-anchored lists to identify neuronal/plasticity modules associated with FS vs BLPM differences.
4. Cross-check directionality at sentinel GC-responsive genes (Fkbp5) and note core limitations.

2. Materials and Methods

2.1 Experimental design

This study re-analysed publicly available chromatin immunoprecipitation sequencing (ChIP-seq) datasets from adult male rat hippocampus under three physiological conditions: baseline early morning (BLAM), baseline late afternoon (BLPM; circadian peak), and 30 min after forced-swim stress (FS). The datasets profiled genome-wide binding of GR, MR and the histone S10, with matched input controls. Each condition included four biological replicates per factor (GR, MR, S10), giving 36 libraries in total. The primary analysis contrasted FS with BLPM to isolate stress-specific binding on a glucocorticoid-high background. The BLAM contrasts (FS vs BLAM; BLPM vs BLAM) were used to define the

scale of trough-to-peak engagement and to contextualise any FS-specific increments seen at 30 min. All processing steps described below were applied uniformly to every factor and condition to maintain comparability.

2.2 Peak calling and preprocessing

Aligned, deduplicated BAM files and matched inputs from the original study were used directly. Peaks were called with MACS2 v2.2.9.1 in paired-end mode (-f BAMPE) against the matched input of the same condition, using the rn6 reference genome and an FDR threshold of ≤ 0.05 . Duplicate reads were retained at the peak-calling stage because deduplication had already been performed upstream by the original authors (1). Standard narrowPeak files were generated per factor and condition (29). Although S10 can appear as broader domains, visual inspection showed punctate enrichments at the time point studied, and using a single peak format for GR, MR and S10 simplified downstream counting and intersection analysis.

2.3 Differential binding analysis

Differential binding was quantified in DiffBind v3.18.0 (DESeq2 backend). Consensus peak sets were defined across conditions at $\geq 4/12$ samples to ensure reproducible occupancy in at least one physiological state while limiting single-replicate artefacts. Read counting used properly paired fragments overlapping each consensus interval. DESeq2 size-factor normalisation and empirical Bayes dispersion estimation were applied peak-wise. Contrasts were FS vs BLPM (primary), FS vs BLAM and BLPM vs BLAM, with positive log₂ fold-change indicating higher signal in the first-named condition. Peaks were called significant at FDR ≤ 0.05 , and extended lists at FDR ≤ 0.05 were generated for pathway analysis (30). Sensitivity to the consensus definition was checked by relaxing the threshold to 2/4 within-condition presence. As expected, relaxing the threshold increased differential peak counts across all contrasts but did not alter the factor rank-order (GR > MR > S10) or the direction of the FS vs BLPM effects. Counts reported in the Results use the stricter $\geq 4/12$ sets. Because BLAM-traversing contrasts may reflect large global shifts in occupancy, FS vs BLPM was interpreted as a GC-high vs GC-high specificity test, while BLAM-traversing contrasts were read as magnitude tests (GC-low $\rightarrow$ GC-high). No spike-in normalisation was available; the same DiffBind/DESeq2 pipeline was applied to all factors and contrasts for consistency.

2.4 Peak annotation and gene association

Differentially bound peaks were annotated with ChIPseeker using TxDb.Rnorsegicus.UCSC.rn6.refGene for genomic coordinates and org.Rn.eg.db for gene symbols. Peaks were assigned to promoter (≤ 3 kb from TSS), exon, intron, UTR, downstream (≤ 3 kb from TES) or distal intergenic classes. Each peak was linked to the nearest gene by linear distance. Where a peak overlapped multiple transcript isoforms of the same gene, the TxDb canonical transcript was used (31). For pathway input, each

differential list was collapsed to a unique set of nearest genes per factor and contrast so one entry per gene even if multiple peaks mapped.

2.5 Data visualisation

Global structure was assessed by principal component analysis (PCA) on variance-stabilised consensus-set counts. PC1/PC2 score plots were used to check state separation and replicate cohesion. MA and volcano plots summarised effect size and significance from the DESeq2 models, with fixed axis limits within each figure to aid comparison of geometries across factors. Heatmaps show z-score-normalised counts per peak (rows) across samples (columns) ordered by condition, highlighting within-state cohesion and between-state differences.

Per-condition occupancy sets were defined for each factor as consensus peaks present in at least one replicate of that factor in that condition. Overlaps between GR, MR and S10 were computed with bedtools intersect using any base-pair overlap and summarised as Venn diagrams. The aim was descriptive visualisation. Locus-level inspection (FKBP5) was carried out in the Integrative Genomics Viewer (IGV) using the provided BAM coverage and peak calls to confirm direction and concordance with summary statistics (32).

2.6 Functional enrichment analysis

Nearest-gene lists derived from differentially bound peaks ($FDR \leq 0.05$) were submitted to Ingenuity Pathway Analysis (IPA; QIAGEN) to place binding differences into biological context. Because inputs derive from ChIP-seq, binding does not equal transcription; therefore, IPA outputs were read as descriptive evidence of pathway presence rather than quantitative claims of activation. Reports were limited to neuronal, cellular and behavioural categories relevant to hippocampal physiology; oncology-labelled pathways were excluded. For canonical pathways, $-\log_{10}(p)$ values are reported in the Results, and activation z-scores are shown when available. Given modest input sizes in some contrasts, especially for FS vs BLPM, z-scores were treated qualitatively to indicate directional concordance across related signalling modules (for example, cAMP/ERK→CREB or RHO/axonal-guidance axes). Where input lists were too small to support a stable profile for MR and S10 in FS vs BLPM, no IPA summary was produced, and those factors were discussed only at the level of binding geometry.

No upstream regulator, causal network or toxicity panels were interpreted, as the aim was to provide compact pathway context for the ChIP-anchored lists. Bar charts rank pathways by $-\log_{10}(p)$ and include z-scores where available to convey pattern-level alignment within the canonical hippocampal plasticity range.

2.7 Quality control and reproducibility checks

Quality checks focused on replicate concordance and symmetry of differential signal. First, PCA of variance-stabilised counts showed clear state separation for GR and moderate

separation for MR, with S10 showing partial overlap, matching the relative magnitudes of differential binding seen in the Results. Second, heatmaps of differentially bound peaks revealed coherent within-state clustering without replicate outliers. Third, relaxing the consensus threshold ($\geq 2/4$ within condition) increased counts but did not change effect direction or the relative ranking of factors, indicating strength to peak-set definition. Together, these checks argue against hidden batch effects or asymmetric library quality between conditions.

2.8 Data availability

All raw and processed ChIP-seq data were obtained from GEO: GSE126706 (Neuro-Epigenetics Research Group, University of Bristol; published by Mifsud et al., Nature Communications 2021, 12:4737. Processed peak files and sample metadata from that study served as inputs for this re-analysis. No new data were generated (1).

2.9 Software and versions

Peak calling used MACS2 v2.2.9.1 in paired-end mode against matched inputs on the rn6 reference genome. Differential binding was performed in DiffBind v3.18.0 with a DESeq2 backend, counting properly paired fragments over consensus peak intervals and applying standard size-factor normalisation and dispersion estimation. Annotation relied on ChIPseeker with TxDb.Rnorvegicus.UCSC.rn6.refGene and org.Rn.eg.db. Interval operations for overlaps used bedtools. Visualisation employed ggplot2, IGV and DiffBind plotting utilities. All analyses were run in R with consistent package versions across factors and contrasts.

3. Results

3.1 Overview

Signal was consistent across replicates and conditions. Principal component analysis (PCA) on consensus peak counts separated samples by physiological state for all factors, most clearly for GR. For GR, PC1 captured $\approx 88\%$ of the variance, with BLAM, BLPM and FS forming discrete clusters. MR showed the same organisation with slightly looser BLPM clustering (PC1 $\approx 80\%$). S10 displayed greater overlap between states (PC1 $\approx 75\%$). Heatmaps of differentially bound (DB) sites recapitulated these patterns: GR clustered tightly by condition; MR showed moderate structure with a narrower correlation range; S10 exhibited weaker segregation, consistent with smaller differential effects at 30 min (Figure 1–2).

3.2 FS vs BLPM

3.2.1 GR

Using the $\geq 4/12$ consensus and DESeq2 in DiffBind, 181 GR peaks were significant at FDR ≤ 0.05 (285 at ≤ 0.10), with a mean $\log_2$ fold change $\approx +1.07$ towards FS. MA and volcano

plots showed a clear right shift and right skew, indicating higher GR occupancy after stress despite both states being glucocorticoid-high (Figure 3A–B). Exemplars included Aspa (+2.29), Abhd11 (+1.96), Relt (+1.90), Usp46 (+1.88) and Esam (+1.84). Fkbp5 was not detected in this contrast, consistent with minimal differential signal at that locus between two GC-high states. Table 3 shows the top 25 GR peaks in FS vs BLPM.

3.2.2 MR

MR yielded 11 significant sites at $FDR \leq 0.05$ (14 at ≤ 0.1), with mean $\log_2$ fold change $\approx +1.31$ towards FS. Top loci mapped near Syt13, Aspa, Nsmf, Relt and Ccne1. These results indicate a smaller FS-related increment for MR relative to GR (Figure 3).

3.2.3 S10

S10 produced 9 sites at $FDR \leq 0.05$ (22 at ≤ 0.10), with mean $\log_2$ fold change ≈ -0.24 and fold changes clustered near zero (Figure 3). Loci included Notum (-0.98), Enpp5 ($+0.71$), Cry1 (-0.69), Nfil3 ($+0.48$) and Egflam (-0.47). These counts and effect sizes indicate limited redistribution of the S10 mark between FS and BLPM at 30 min.

3.2.4 IPA (GR)

Where list sizes permitted, peak-anchored genes from FS_vs_BLPM were submitted to Ingenuity Pathway Analysis (IPA). For GR, the extended list ($FDR \leq 0.05$) retrieved CREB signalling in neurones, GPCR cascades (including dopamine–DARPP32), NF1/RAS–MAPK, PI3K/AKT with predicted PTEN inhibition, RHO GTPase/cytoskeleton, myelination, and circadian rhythm signalling. Additional stress-linked components were present, including GADD45/senescence and IFN- γ /JAK–STAT modules. MR and S10 lists for this contrast were too small for stable inference and are not shown (Figure 6). MR and S10 lists for this contrast were too small for stable inference and are not shown

3.2.5 Counts and checks

Across the $\geq 4/12$ consensus, FS_vs_BLPM was compact and consistent for GR (181 at $FDR \leq 0.05$; 285 at ≤ 0.1) and modest for MR (11; 14) and S10 (9; 22), spanning promoter, intronic and distal contexts. Relaxing the consensus to 2/4 increased counts as expected but preserved the factor rank-order, so results are presented at the stricter threshold. The size of the tested search space per factor is shown in Table 2.

3.3 BLAM contrasts

3.3.1 FS vs BLAM

From the hormone trough, both receptors show strong induction. GR yielded 1,114 significant sites ($FDR \leq 0.05$; mean $\log_2$ fold change $\approx +4.15$), exemplified by Relt; MR yielded 769 ($\approx +2.36$), exemplified by Aspa; S10 changes were modest (150 sites). MA/volcano plots show pronounced right shifts for the receptors (Figure 4, count in Table 1).

3.3.2 BLPM vs BLAM

Circadian engagement also drove large receptor gains. GR yielded 1,062 sites ($\approx +3.44$), exemplified by Arhgap31; MR 809 ($\approx +2.16$), exemplified by Lamb2; S10 changes were small (26 sites). These contrasts establish the scale of trough-to-peak recruitment.

3.4 FKBP5

FKBP5 appeared in GR lists whenever BLAM was the comparator but not in GR FS_vs_BLPM, consistent with FKBP5 reporting GC availability (trough→peak) rather than distinguishing stress from the circadian peak at 30 min.

3.5 Overlaps

Condition-wise overlaps ($\geq 4/12$) showed widespread S10 with nested MR/GR and a larger triple intersection under GC-high states: $GR \cap MR \cap S10$ rose from 126 at BLAM to 218 at BLPM and remained 218 after FS. This indicates that circadian phase expands a receptor-compatible, S10-rich search space into which stress adds selective GR occupancy (Figure 5).

3.6 IPA

To place differential receptor binding into biological context, peak-nearest gene lists (DB FDR ≤ 0.05) were submitted to IPA.

FS_vs_BLPM Canonical pathways concentrated on hippocampal plasticity signalling. Top terms by $-\log_{10}(p)$ included CREB Signalling in Neurones ($-\log_{10} p = 4.54$; $z = +1.34$), Axonal Guidance Signalling (4.36; +1.15), Dopamine–DARPP32 Feedback in cAMP Signalling (3.99; +1.63), G-Protein Coupled Receptor Signalling (3.97; +1.00), ERK/MAPK Signalling (3.89; +1.86), PTEN Signalling (3.86; -0.45), cAMP-mediated signalling (3.35; +2.00) and Circadian Rhythm Signalling (3.80; +1.13). “Diseases & Functions” mirrored this with behaviour-linked annotations, including Cognition ($p = 6.46 \times 10^{-15}$; $z = +2.39$), Learning (8.84×10^{-14} ; +2.34) and Synaptic transmission (3.49×10^{-12} ; +2.13). Overall, the profile matches the peak-level result, a focused, GR-centred increment on a circadian engaged background, linking neuromodulatory GPCR input to cAMP/ERK→CREB and cytoskeletal control, with a modest circadian component (Figure 6).

BLAM-traversing contrasts. For FS_vs_BLAM, the same plasticity kernel appeared at higher significance and larger z-scores, e.g., CREB Signalling in Neurones ($-\log_{10} p = 9.61$; $z = +3.32$), ERK/MAPK Signalling (9.19; +3.49), Axonal Guidance (8.73; +2.36), Myelination Signalling (7.52; +2.58), PTEN Signalling (7.37; +2.24) and cAMP-mediated signalling (7.28; +3.11). For BLPM_vs_BLAM, magnitudes were smaller than FS_vs_BLAM but still strong, CREB Signalling (7.15; +2.56), ERK/MAPK (6.70; +2.69), Axonal Guidance (6.29; +2.10), Dopamine–DARPP32 Feedback (6.13; +2.06) and Circadian Rhythm Signalling (4.55; +1.00). These context contrasts frame the primary result, moving from trough to GC-high broadly engages plasticity modules; FS_on_BLPM recovers a compact subset of the same machinery.

MR and S10. For FS_vs_BLPM there were too few MR and S10 targets for a stable IPA profile, so no bar plots are shown for the primary contrast. In the BLAM-traversing contrasts, MR returned overlapping repertoires at lower amplitude than GR, consistent with the DiffBind results. S10 outputs were modest and consistent with a broadly permissive chromatin state rather than strong mark-specific direction at 30 min.

Across contrasts, IPA re-identified a conserved hippocampal plasticity backbone, GPCR to cAMP/PKA to ERK/MAPK to CREB, PI3K/AKT with PTEN involvement, RHO GTPase/cytoskeleton, Axonal Guidance and Myelination, with an explicit Circadian component, and compact stress-biased additions when FS is contrasted directly with BLPM. Because FS_vs_BLPM gene counts are smaller than the usual 500–2000-gene window for IPA, bar charts are ranked by $-\log_{10}(p)$ with accompanying z-scores, and direction is interpreted qualitatively from these ChIP-anchored lists.

4. Discussion

This project investigated receptor–genome interactions of GR, MR and the histone mark S10 under circadian and stress conditions in rat hippocampus. The main question was whether an acute stressor generates a distinct genomic footprint when the system is already at the daily GC peak (33). FS vs BLPM at 30 min was treated as the primary comparison. The key findings are consistent across analyses. In FS vs BLPM, GR showed a clear right-shifted distribution with 181 differentially bound peaks at $FDR \leq 0.05$ (285 at ≤ 0.10 ; mean $\log_2FC \approx +1.07$). MR changed at far fewer sites (11–14) but in the same direction, while S10 differed at only 9–22 loci with fold-changes clustered near zero (mean ≈ -0.24). In the BLAM contrasts, both receptors showed large gains thousands of GR sites; hundreds of MR sites, and S10 remained modest. PCA and heatmaps mirrored these results; GR separated cleanly by state, MR separated more weakly, and S10 showed greater overlap. FKBP5 behaved as expected for a GC sentinel, it appeared whenever BLAM was the comparator but not in GR FS vs BLPM, which supports the interpretation that it reports trough to peak hormone availability rather than distinguishing two GC-high states at 30 min (34). These patterns together indicate that the circadian peak primes a broad receptor-competent chromatin landscape and that stress on this background adds a selective, GR-centred increment (35).

The GR result in FS vs BLPM is selective rather than global, and the plot patterns support this. MA/volcano plots show a right-shift and right-skew, consistent with increased GR occupancy on top of a GC-high baseline. The mean effect is moderate, which is exactly what we would expect if BLPM has already engaged much of the GR-competent genome (34). The BLAM contrasts supply the standardisation. From the trough, GR gains are very large (1,114 sites for FS vs BLAM; 1,062 for BLPM vs BLAM; mean $\log_2FC \approx +4.15$ and $+3.44$), so

a 181-site increment at 30 min on BLPM should be read as targeted rather than weak. Loci in the FS vs BLPM set span promoter-proximal (Aspa, Esam), intronic (Relt) and distal intergenic (Abhd11, Usp46) regions, which is the mixed architecture expected for in vivo GR action in hippocampus (1). This pattern fits established chromatin principles. Most inducible GR binding occurs at regions that are already accessible before stimulation; only a minority of sites open de novo (36,37). Baseline accessibility and motif environment therefore constrain where GR can increase, so when many sites are competent at BLPM, a stress pulse elevates occupancy at a subset of those elements instead of redefining the whole landscape. Foundational work demonstrated that cell- and context-specific accessibility pre-determines GR occupancy patterns, and that pulsed hormone inputs generate spatial-temporal binding different from constant exposure (38,39). Taken together, those studies predict a selective FS on BLPM increment at a snapshot time point.

The MR result in FS vs BLPM is sparse but informative. Only 11–14 sites pass FDR, although the mean effect per site is like GR and directionally aligned. This differs from the BLAM contrasts, where MR shows hundreds of changes with mean $\log_2FC \approx +2.2-2.4$. This could be because the system saturates at the circadian peak. MR has higher affinity for corticosterone than GR, so it's already mostly occupied at baseline and is pushed toward saturation by the circadian rise (40,41). By late afternoon, many MR-competent sites are likely near maximal occupancy, leaving little headroom for an additional increment at 30 min after stress (42). This fits the established division of labour in which MR sets tonic/circadian tone and GR supplies larger phasic adjustments to acute changes. Sampling time may also contribute. MR and GR differ in kinetic properties and subcellular localisation; if MR's dynamic response peaks before or after 30 min, a single snapshot could underestimate it. Bulk hippocampus then averages different subfields and cell types, which can dilute localised differences (43,44). Together these points could explain why MR contributes strongly when crossing BLAM yet adds little on top of BLPM at 30 min.

S10 shows very limited redistribution in FS vs BLPM. Only 9–22 sites differ and fold-changes cluster near zero. Separation by PCA is weaker than for the receptors, and the context contrasts also show modest S10 movement compared with GR/MR. Here, S10 looks like an enabling mark, it makes sites responsive but doesn't drive the differences between the two GC-high states (45). A few small changes near clock genes (Cry1, Nfil3) suggest minor circadian alignment, but the overall effect is modest (46). This is compatible with known kinetics, phosphorylation–acetylation of H3 in adult hippocampus after psychological stress can be rapid and transient, with a clear link to ERK/MSK signalling and immediate-early gene induction in dentate gyrus (47,48). In that framework, a short S10 pulse could occur early after FS and decline by 30 min, while GR occupancy at some sites remains elevated. Alternatively, BLPM may already set S10 at high levels, leaving little room for further change.

The per-condition overlap analysis supports such a model, the GR $\cap$ MR $\cap$ S10 intersection expands from 126 regions at BLAM to 218 at BLPM and stays at 218 after FS, while GR still adds a selective increment within that space.

Reading across the three physiological states, there is a consistent pattern. From BLAM to BLPM, circadian engagement seems strong, with MR and GR both increasing and the MR–GR–S10 triple overlap expanding. When FS is added on top of BLPM and measured at 30 min, the overall S10-positive search space looks stable. MR contributes little extra, while GR shows a targeted increase. FKBP5's behaviour fits this view. It is present whenever BLAM is the comparator but not in GR FS vs BLPM, which supports the interpretation that FKBP5 signals glucocorticoid availability rather than distinguishing between two GC-high states at a single acute time point. This internal check reduces the risk that our main contrast simply re-identified classic GC targets and supports the idea that the analysis captured a context-dependent subset at the peak.

Timing is also crucial for interpretation. The HPA axis releases glucocorticoids in ultradian pulses on top of the circadian rhythm (33, 38). Pulses drive cyclic GR-dependent transcription and produce chromatin accessibility and long-range interaction patterns that differ from constant hormone. A single 30 min sample can catch only one phase of this cycle, so chromatin marks or cofactor states may have already risen and fallen even while GR binding remains detectable at some sites. This can explain the compact GR FS vs BLPM set and the near-zero S10 differences. A brief S10 surge may have occurred earlier, with GR occupancy still elevated at a subset of primed loci at 30 min. An MR transient could also fall outside this window, contributing to the sparse MR list despite strong trough-crossing changes in the BLAM contrasts. These timing points are consistent with well-cited studies of GR dynamics under pulsatile stimulation (38,49,50).

Pathway analysis helps place the GR subset added by FS in context. Even with a small input set, Ingenuity Pathway Analysis pointed to hippocampal plasticity signalling, including CREB signalling in neurones, cAMP-mediated signalling, ERK/MAPK cascades, GPCR pathways such as dopamine–DARPP32 feedback, PI3K/AKT with a small mixed PTEN signal, RHO GTPase and cytoskeleton remodelling, axonal guidance and myelination, and a circadian term (52). Behavioural functions (cognition, learning, synaptic transmission) showed positive activation scores; however, because the input is ChIP-anchored and the list is small, These are interpreted as descriptive rankings rather than quantitative claims. The same pathway backbone appears with greater magnitude in the BLAM-crossing contrasts, which supports a model of circadian priming followed by selective GR addition at 30 min during FS. This profile is consistent with prior work showing that corticosteroid receptors integrate neuromodulator input and kinase cascades to shape hippocampal plasticity and behaviour (53). It also aligns with genome-wide studies reporting overlapping but distinct receptor

programs, with MR emphasising tonic and circadian organisation and GR contributing stronger, dynamic responses.

Two locus-level observations support the pathway results. First, the GR peaks added by FS occur at promoters, introns, and distal regions. This fits a mixed promoter–enhancer model in which pre-existing accessibility and 3D topology allow rapid GR recruitment (36, 54). Studies in other systems show that GR usually binds sites that are already open, with only limited de novo opening, and that pulsed hormone creates distinct DNase-accessibility patterns with different lifetimes (36, 50). This offers a clear mechanism for the compact GR increase seen here. Second, several FS-added loci, such as *Relt* and *Usp46*, are plausible regulators of stress signalling or synaptic control, which aligns with the IPA recovery of plasticity and cytoskeletal pathways. A focused motif analysis of the GR-added summits for GREs together with AP-1 or CRE motifs would test this idea and tighten the link to the neuromodulator–CREB axis (56).

The MR pattern across conditions also matches the physiology. With its high affinity, MR is substantially occupied at baseline and rises further during the circadian increase; this is what the BLAM contrasts show in the results. The lack of a large MR increment at 30 min on BLPM is therefore consistent with a ceiling effect. Studies consistently describe complementary roles for MR and GR in hippocampus. These findings give a genomic footprint for that division in a comparison that holds both sides GC-high and looks for added specificity from stress at one acute time point.

The S10 result fits prior hippocampal stress work. Psychological stress, including forced swim, can rapidly trigger H3 phospho-acetylation in the dentate gyrus through ERK/MSK signalling, with tight links to immediate-early gene induction and behaviour. These events are transient, so a single 30 min sample may reasonably show a permissive profile rather than a net difference between two glucocorticoid-high states. The per-condition overlaps also suggest a permissive ceiling at BLPM. S10 marking expands from BLAM to BLPM and then remains broad after FS, while GR selects within this S10-rich set (57).

FKBP5's pattern across contrasts works as an internal check. It appears whenever BLAM is the comparator but is absent from GR FS vs BLPM. That fits its role as a glucocorticoid-responsive chaperone that tracks hormone availability and GR sensitivity. Reviews describe strong regulation by stress and glucocorticoid exposure and well-documented gene-by-environment interactions in humans (58, 59). This helps explain why FKBP5 separates trough from peak but does not necessarily distinguish two GC-high states at a single acute time point in hippocampus.

Several methodological choices affect interpretation and should be stated clearly. First, all measurements were taken at 30 min. Very early transients (for example a fast S10 pulse) and later secondary phases were not captured. Because the endocrine system is pulsatile,

the phase of sampling can influence genomic read-outs, which adds uncertainty to negative or near-zero findings for marks with fast kinetics (60). Second, the tissue is bulk hippocampus, so subfield- or cell-type-specific responses can be diluted (61). Third, nearest-gene anchoring is conservative and ignores three-dimensional chromatin, so distal enhancer–promoter assignments may be wrong for some peaks. Fourth, statistical power differs by factor; GR provided enough FS vs BLPM sites for descriptive pathway analysis, whereas MR and S10 did not. Fifth, A strict consensus threshold was used ($\geq 4/12$), which improves reliability at some cost to sensitivity; relaxing to 2/4 increased counts but did not change the rank ordering across factors. Finally, Consensus peak counts were analysed in DiffBind using DESeq2's count-based scaling. A large global shift in occupancy could, in principle, affect such scaling. The main interpretive point is the relative compactness of FS vs BLPM compared with the BLAM contrasts, and that pattern holds under the analysis choices used throughout this project.

Within these boundaries, the data answer the original hypotheses directly. First, GR retains inducible capacity at 30 min on a GC-high background; MR moves in the same direction but adds far fewer sites; S10 is flat between two GC-high states. Second, S10 behaves permissively, consistent with a broad S10-positive site at BLPM and the possibility of a brief, earlier transient. Third, co-occupancy is common and increases under GC-high states; the triple $GR \cap MR \cap S10$ intersection rises strongly from BLAM to BLPM and stays high after FS. Fourth, pathway context for FS vs BLPM collapses to a conserved plasticity kernel with small stress-biased additions, rather than revealing a novel network.

These conclusions have clear implications and point to concrete next steps that would strengthen the mechanistic links. A short time-course centred on BLPM for example 5, 15, 30, 60, 120 minutes, for GR, MR and S10 would determine whether S10 or MR respond outside the 30 min window. This matters in an ultradian system and would convert a single snapshot into a short time-course. Adding matched transcriptional read-outs in the same samples for example, RNA-seq with intronic counts to capture nascent change, would identify which FS-added GR peaks lead to expression differences at the same time points and would test whether the cAMP/ERK to CREB and cytoskeletal axes predicted by IPA show directional movement. A focused motif analysis at FS-added GR summits would test enrichment for GREs combined with AP-1 or CRE motifs and connect selective recruitment to plausible co-factor usage. Subfield- or cell-type-resolved assays for example, CUT&Tag in sorted neuronal populations or single-nucleus approaches, would identify where in hippocampus the GR increment occurs and whether MR and S10 differ across regions. For a small number of representative loci (*Aspa*, *Esam*, *Relt*, *Abhd11*, *Usp46*), targeted assays could test enhancer–promoter coupling to improve gene assignment.

Although this study was limited to rat hippocampus and did not include behavioural tests, the pattern still speaks to timing-dependent stress biology. We see a circadian primed, S10-rich landscape, with a targeted GR increase after acute stress. Prior work shows that the timing and context of corticosteroid exposure shape hippocampal plasticity (2). Fkbp5 behaves in line with this idea, it is sensitive at the trough but relatively insensitive at the GC peak, which fits gene-by-environment models of stress responsivity (7,8). Receptor mapping in hippocampus also links MR and GR programs to neuroplasticity modules relevant to stress-related vulnerability (9). These findings lead me to suggest follow-up studies that test stress effects with time of day in mind.

In summary, contrasting FS with BLPM isolates a selective GR-centred addition on a circadian engaged background. MR moves in the same direction but adds little at 30 min, and S10 behaves permissively between two GC-high states in bulk hippocampus. The pathway context reduces to a familiar hippocampal plasticity stack with a circadian component. Alongside the BLAM contrasts, these findings provide a phase-sensitive account of corticosteroid receptor action in hippocampus.

5. Conclusion

At the circadian glucocorticoid peak, acute stress leaves a selective, GR-centred genomic footprint in rat hippocampus. On a broadly permissive, S10-rich background with extensive MR/GR occupancy, stress recruits GR at a focused subset of loci while MR contributes little additional binding and S10 shows only small differences at 30 min. Pathway context converges on neuromodulatory GPCR→cAMP/ERK→CREB signalling with cytoskeletal and axon-guidance support and a modest circadian component. Methodologically, a GC-high vs GC-high comparator, a conservative consensus-peak framework and internal sentinels such as FKBP5 support these inferences; limits of a single time point, bulk tissue and nearest-gene anchoring temper mechanism but not direction. Circadian phase gates receptor access, so the comparator determines whether the analysis measures trough-to-peak magnitude or stress-specific increments at the peak (FS vs BLPM). Phase-aware kinetics combined with matched RNA-seq and focused motif analysis are logical next steps to connect selective occupancy to transcription and mechanism.

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Figures and Tables

Factor	Contrast	Significant_FDR_0.05	Significant_FDR_0.10	Up_FDR_0.05	Down_FDR_0.05	Mean_log2FC_FDR_0.05	Median_log2FC_FDR_0.05
GR	FS vs BLPM	186	297	186	0	1.07	1.03
GR	FS vs BLAM	1304	1312	1301	3	4.14	3.91
GR	BLPM vs BLAM	1242	1260	1241	1	3.41	3.19
MR	FS vs BLPM	11	14	11	0	1.31	1.32
MR	FS vs BLAM	872	902	872	0	2.29	2.02
MR	BLPM vs BLAM	924	967	922	2	2.11	1.9
S10	FS vs BLPM	9	22	2	7	-0.24	-0.28
S10	FS vs BLAM	160	363	67	93	0.14	-0.2
S10	BLPM vs BLAM	28	34	22	6	0.57	0.72

Table 1. Differential binding summary for GR, MR and S10 across contrasts

Counts of differentially bound peaks for FS vs BLPM, FS vs BLAM, and BLPM vs BLAM.

Significant_FDR_0.05 and Significant_FDR_0.10 give the number of sites at $FDR \leq 0.05$ and ≤ 0.10 . Up_FDR_0.05 and Down_FDR_0.05 give the direction at $FDR \leq 0.05$.

Mean_log2FC_FDR_0.05 and Median_log2FC_FDR_0.05 are calculated over significant sites only. Positive log2FC indicates higher signal in the first-named condition of the contrast. Values are from DiffBind with DESeq2 normalisation and a consensus peakset built at minOverlap = 4.

Factor	Consensus peaks
GR	1385
MR	1210
S10	40544

Table 2. Consensus peak counts for GR, MR and S10

Number of consensus peaks per factor compiled across BLAM, BLPM and FS. Consensus peaks were defined from dba.count with summits = 200 and minOverlap = 4, using all samples for that factor. These counts describe the analysed search space used for differential testing.

chr	start	end	log2FC	FDR	SYMBOL	annotation
chr10	59884966	59885366	2.29	2.42E-09	Aspa	Promoter (2-3kb)
chr12	24733195	24733595	1.96	2.42E-09	Abhd11	Distal Intergenic
chr1	165839829	165840229	1.9	1.15E-08	Relt	Intron (NM_001108494/361614, intron 2 of 8)
chr14	36789505	36789905	1.88	0.0000136	Usp46	Distal Intergenic
chr8	39998308	39998708	1.84	0.0000053	Esam	Promoter (<=1kb)
chr6	110704666	110705066	1.79	0.000112	Lrrc74a	Intron (NM_001037783/314328, intron 1 of 14)
chr16	56386932	56387332	1.75	0.000135	Tusc3	Intron (NM_001004212/290783, intron 8 of 9)
chr6	124701390	124701790	1.71	1.15E-08	Rps6ka5	Intron (NM_001108048/314384, intron 1 of 15)
chr12	13853353	13853753	1.68	3.66E-08	Tnrc18	Intron (NM_001107123/304302, intron 4 of 34)
chr8	68928011	68928411	1.56	0.00124	Smad6	Intron (NM_001109002/367100, intron 3 of 3)
chr17	70655709	70656109	1.48	0.00147	Pfkfb3	Distal Intergenic
chr1	250750433	250750833	1.45	0.00132	Asah2	Intron (NM_181386/353229, intron 5 of 9)
chr12	50366540	50366940	1.4	0.00332	Tpst2	Intron (NM_001008508/288719, intron 1 of 6)
chr3	2251731	2252131	1.4	0.0000103	Nsmf	Intron (NM_144738/246246, intron 25 of 34)
chr10	16942569	16942969	1.38	0.00222	Dusp1	Distal Intergenic
chr2	33060103	33060503	1.37	0.00222	Cd180	Distal Intergenic
chr6	59868497	59868897	1.37	0.00343	Arl4a	Distal Intergenic
chr5	149053668	149054068	1.37	0.00343	Laptm5	Intron (NM_053538/89783, intron 1 of 7)
chr1	221283928	221284328	1.37	0.0025	Cdc42ep2	Promoter (2-3kb)
chr1	87001040	87001440	1.36	0.00466	Hnrnp1	Intron (NM_001134760/80846, intron 6 of 11)
chr12	24836219	24836619	1.34	0.00346	Mettl27	Distal Intergenic
chr17	46727815	46728215	1.33	0.00132	Elmo1	Intron (NM_001108415/361251, intron 5 of 21)
chr15	13512057	13512457	1.32	0.000639	Ptpg	Intron (NM_134356/171357, intron 1 of 41)
chr16	7884269	7884669	1.32	0.00631	Btd	Distal Intergenic
chr5	142216575	142216975	1.31	0.00708	RGD1563049	Distal Intergenic

Table 3. Top 25 GR peaks higher in FS than BLPM

Annotated list of the 25 GR peaks with the largest positive log2FC in FS vs BLPM at FDR ≤ 0.05. Peaks are ranked by log2FC and then FDR. Columns report genomic coordinates (rn6), log2FC, FDR, nearest gene symbol, and annotation class from ChIPseeker. Positive log2FC indicates higher GR binding in FS than BLPM.

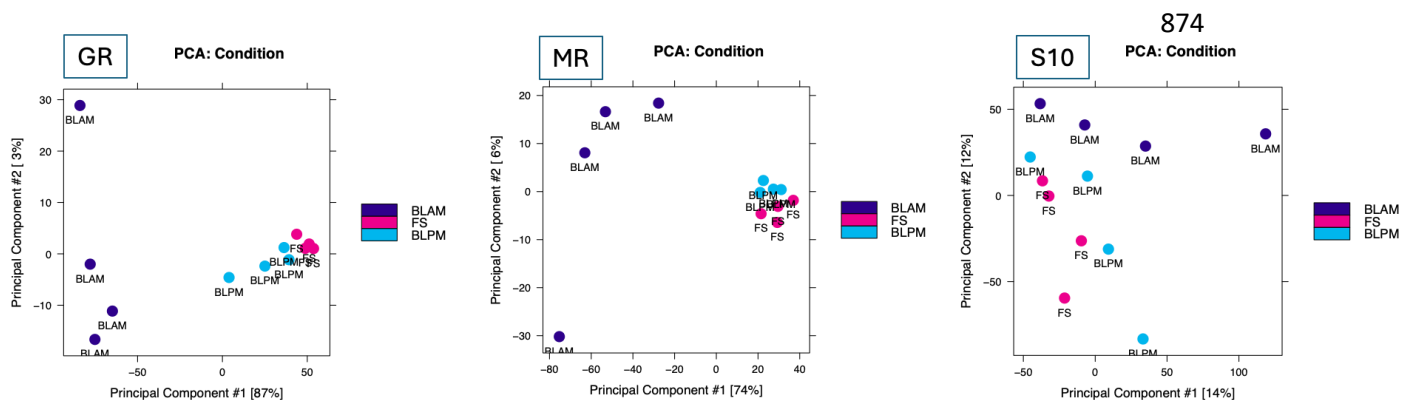


Figure 1. Principal component analysis of consensus peak counts

PC1 separates physiological state for all factors. GR shows the strongest separation (PC1 $\approx$ 88%), MR is moderate ($\approx$ 80%), and S10 shows partial overlap ($\approx$ 75%). BLAM, BLPM and FS form distinct clusters most clearly for GR. Points are coloured by condition; each point is one replicate.

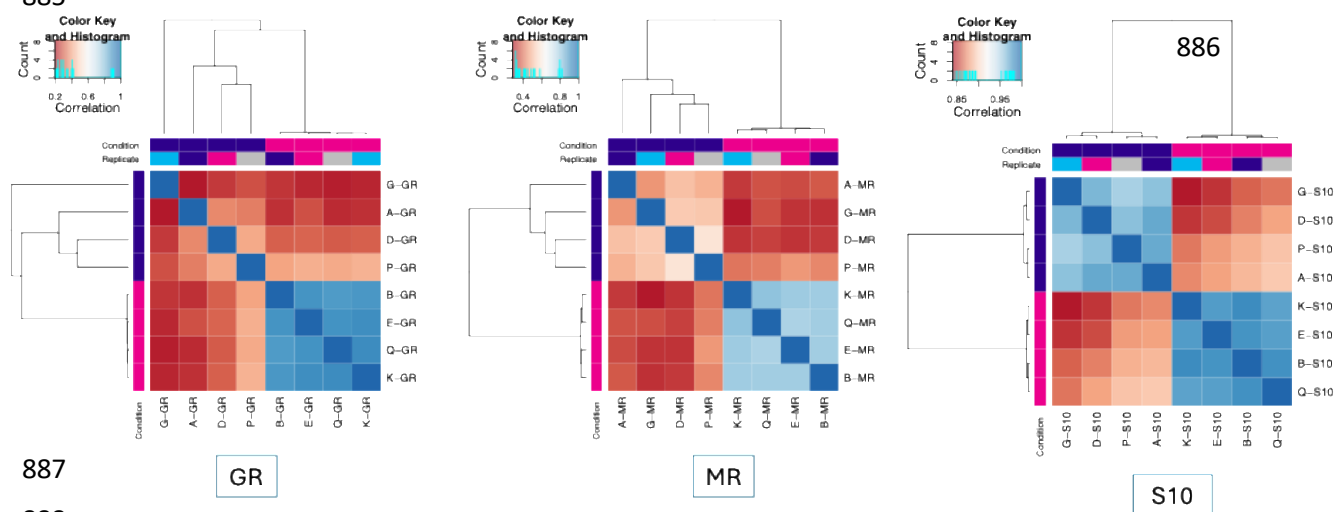


Figure 2. Heatmaps of differentially bound sites across conditions

Z-score heatmaps (rows = DB peaks, columns = samples) ordered by condition. GR clusters tightly by state, MR shows a narrower correlation range, and S10 segregation is weaker. Patterns match the PCA and reflect smaller differential effects at 30 minutes for S10.

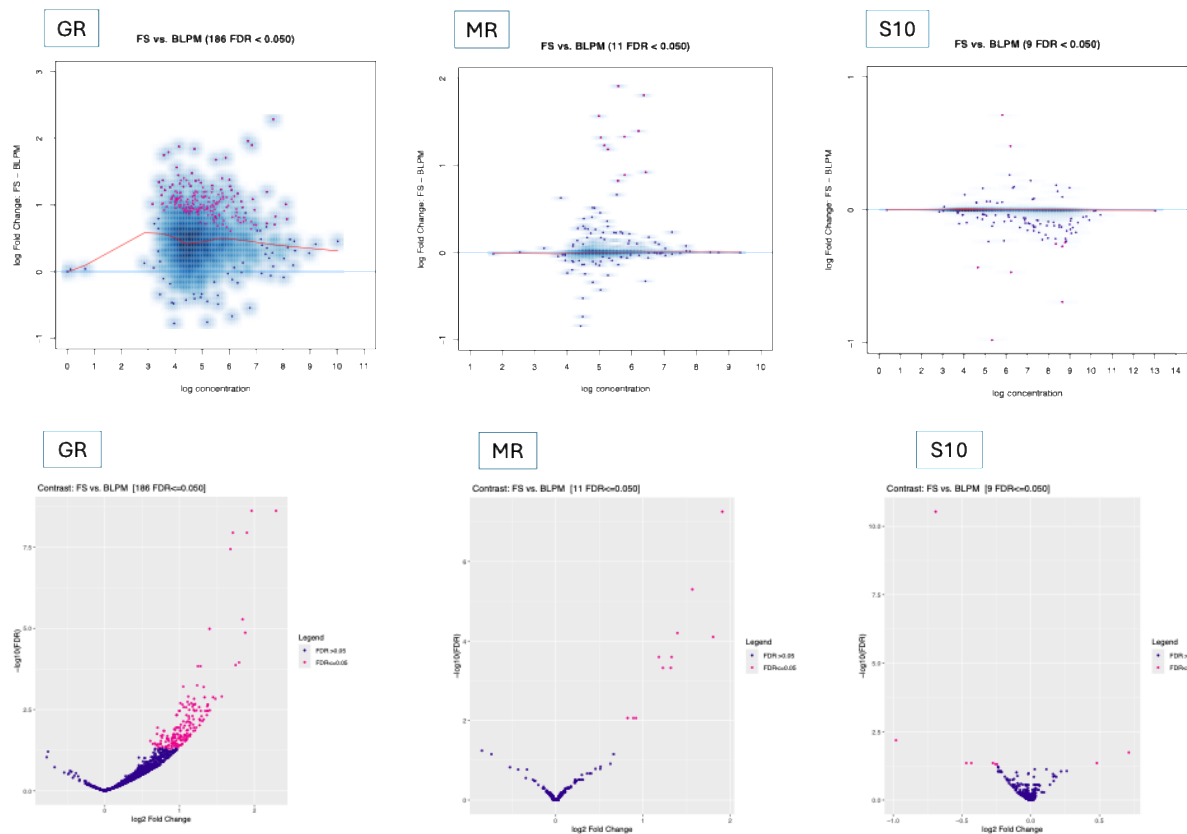


Figure 3. FS vs BLPM at 30 minutes: MA and volcano plots for GR, MR and S10

Differential binding was tested on $\geq 4/12$ consensus peaks using DESeq2. GR shows 181 significant peaks at $FDR \leq 0.05$ (285 at ≤ 0.1 ; mean $\log_2FC \approx +1.07$). Example GR-linked loci include Aspa (+2.29), Abhd11 (+1.96), Relt (+1.90), Usp46 (+1.88) and Esam (+1.84). MR shows 11 significant sites (14 at ≤ 0.1 ; mean $\log_2FC \approx +1.31$). S10 shows 9 significant sites (22 at ≤ 0.1 ; mean $\log_2FC \approx -0.24$) with fold-changes near zero; examples include Notum (-0.98), Enpp5 (+0.71), Cry1 (-0.69), Nfil3 (+0.48) and Egflam (-0.47). Axes are fixed across panels to allow direct comparison.

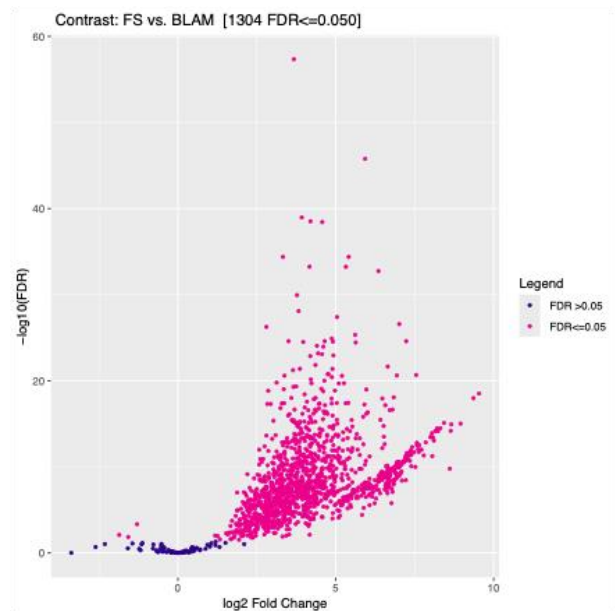
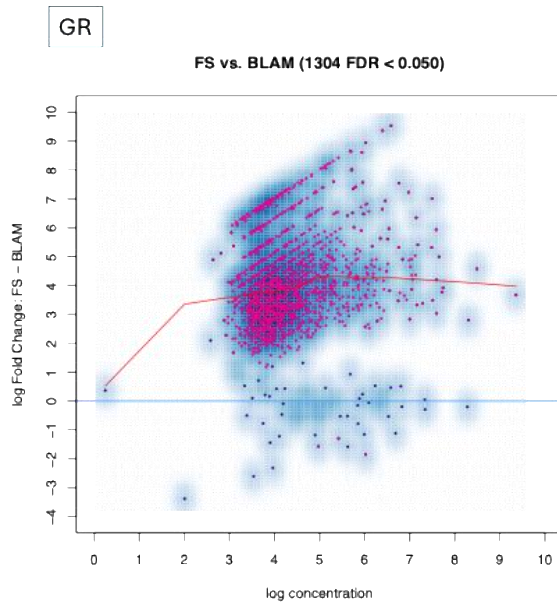
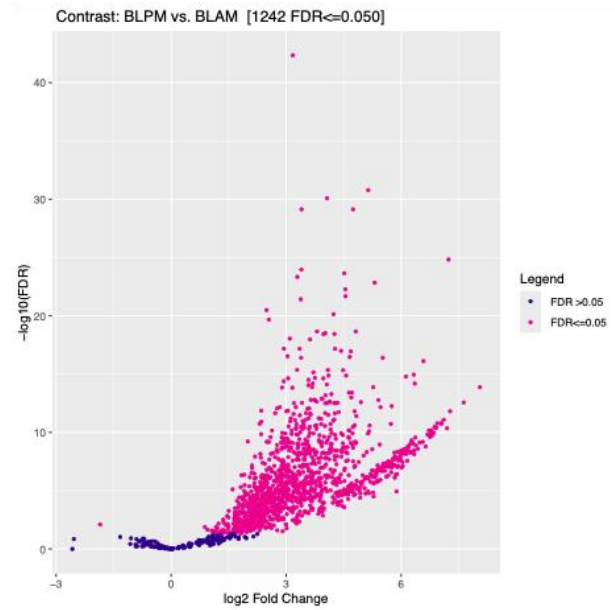
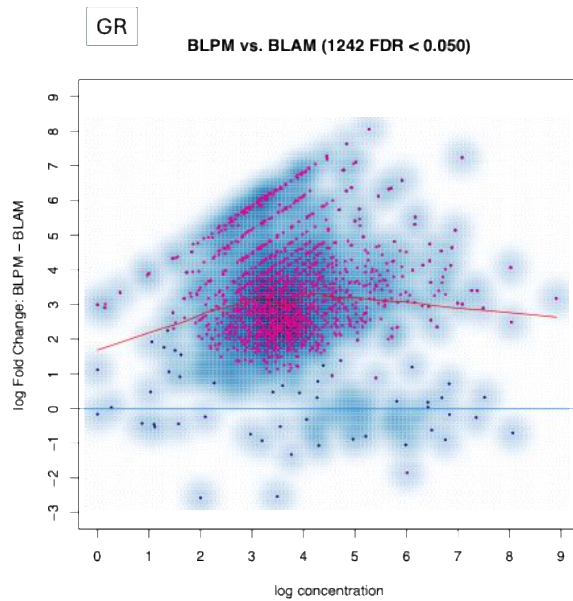


Figure 4. BLAM-crossing contrasts to set scale: FS vs BLAM and BLPM vs BLAM

MA and volcano plots for GR, MR and S10 using the same pipeline as Figure 3. From the trough, GR shows large gains (FS vs BLAM: 1,114 sites; BLPM vs BLAM: 1,062 sites; mean $\log_2\text{FC} \approx +4.15$ and $+3.44$). MR also gains strongly (769 and 809 sites; mean $\approx +2.36$ and $+2.16$). S10 changes are modest (150 and 26 sites). These contrasts define the magnitude of trough-to-peak recruitment used to interpret the selective FS-on-BLPM increment.

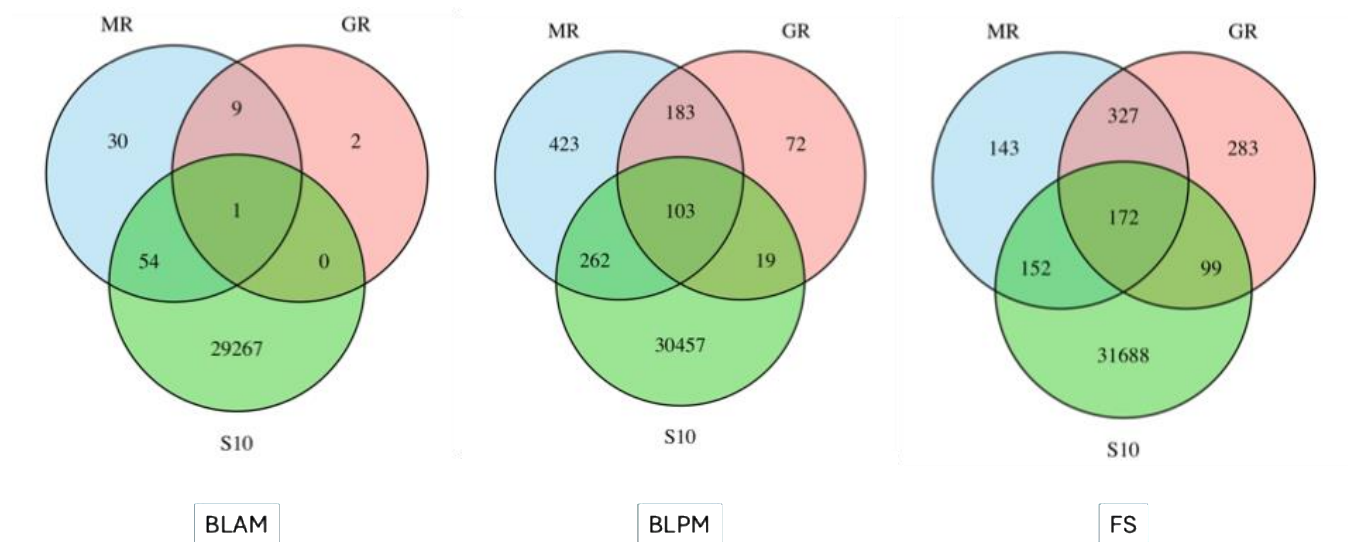
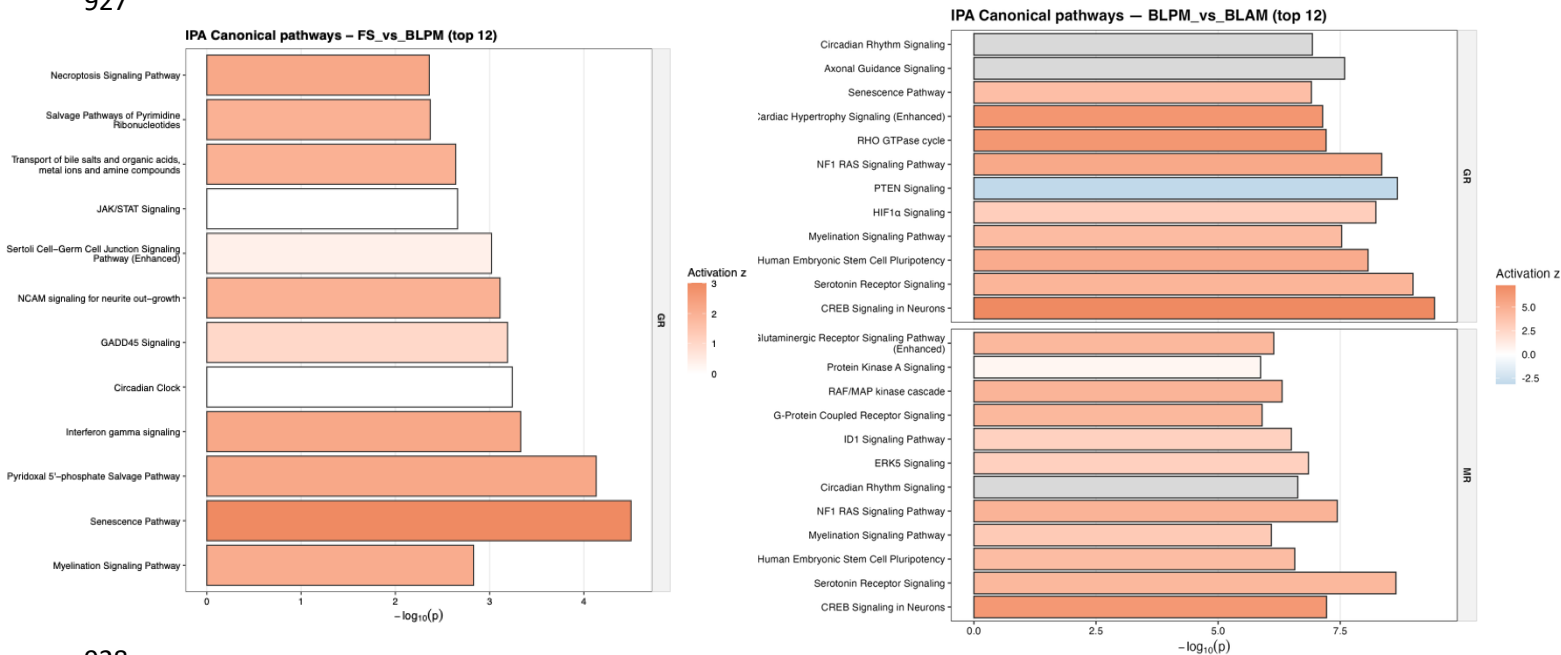


Figure 5. Condition-wise overlaps of GR, MR and S10 occupancy

Venn diagrams per condition based on per-condition occupancy sets ($\geq 4/12$ consensus; any base-pair overlap; descriptive only). The triple intersection ($GR \cap MR \cap S10$) increases from 126 regions at BLAM to 218 at BLPM and remains 218 after FS. This indicates an S10-rich, receptor-compatible space under GC-high states, within which FS adds selective GR occupancy.

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Figure 6. Ingenuity Pathway Analysis for GR nearest-gene lists

Canonical pathway rankings ($-\log_{10} p$) with activation z-scores where available. FS vs BLP (extended list, $FDR \leq 0.1$) recovers a compact plasticity set, including CREB Signalling in Neurones (4.54; $z = +1.34$), Axonal Guidance (4.36; +1.15), Dopamine–DARPP32 Feedback (3.99; +1.63), ERK/MAPK (3.89; +1.86), PTEN (3.86; -0.45), cAMP-mediated signalling (3.35; +2.00) and Circadian Rhythm Signalling (3.80; +1.13). Behaviour-linked functions (Cognition, Learning, Synaptic transmission) show positive scores. The same pathways appear with larger magnitudes in FS vs BLAM and BLP vs BLAM. MR and S10 were not plotted for FS vs BLP due to small input lists.